CompSci 234/NetSys 210 Advanced Topics in Networking

Winter 2019

Unit 9: Peer-to-Peer Networks

Cheng-Hsin Hsu (chsu@cs.nthu.edu.tw)

Slide adopted from Prof. Venkatasubramanian's and Kurose/Ross' book materials

Agenda

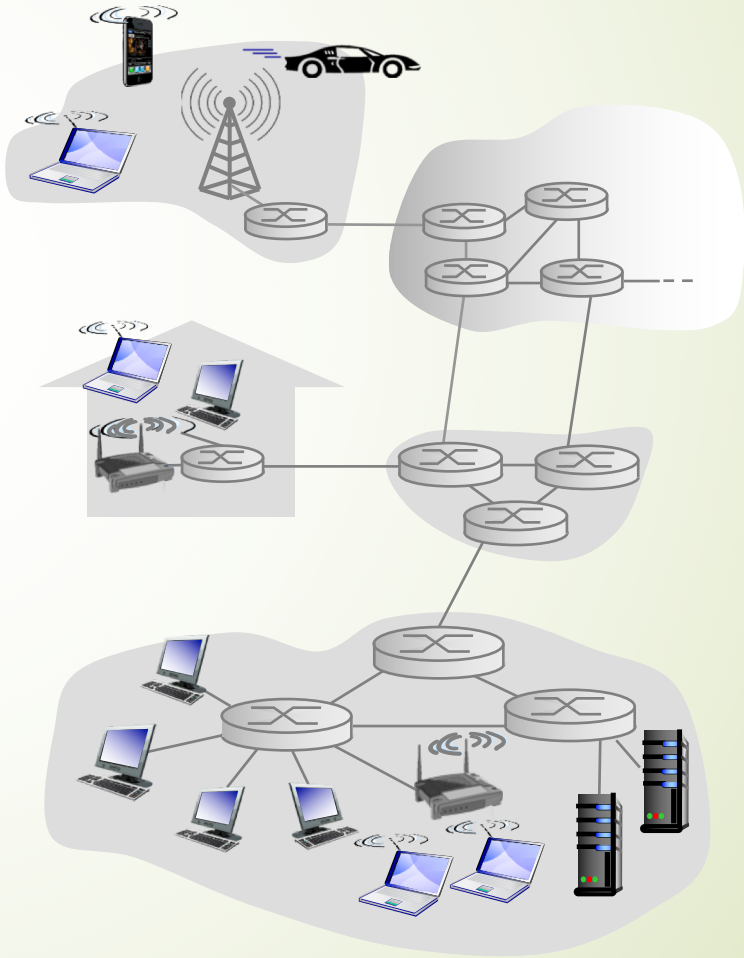
- Overview on P2P Networks
- P2P Applications
- Properties of P2P Systems
- Structured and Unstructured P2P
- Distributed Hash Table (DHT)

Pure P2P Architectures

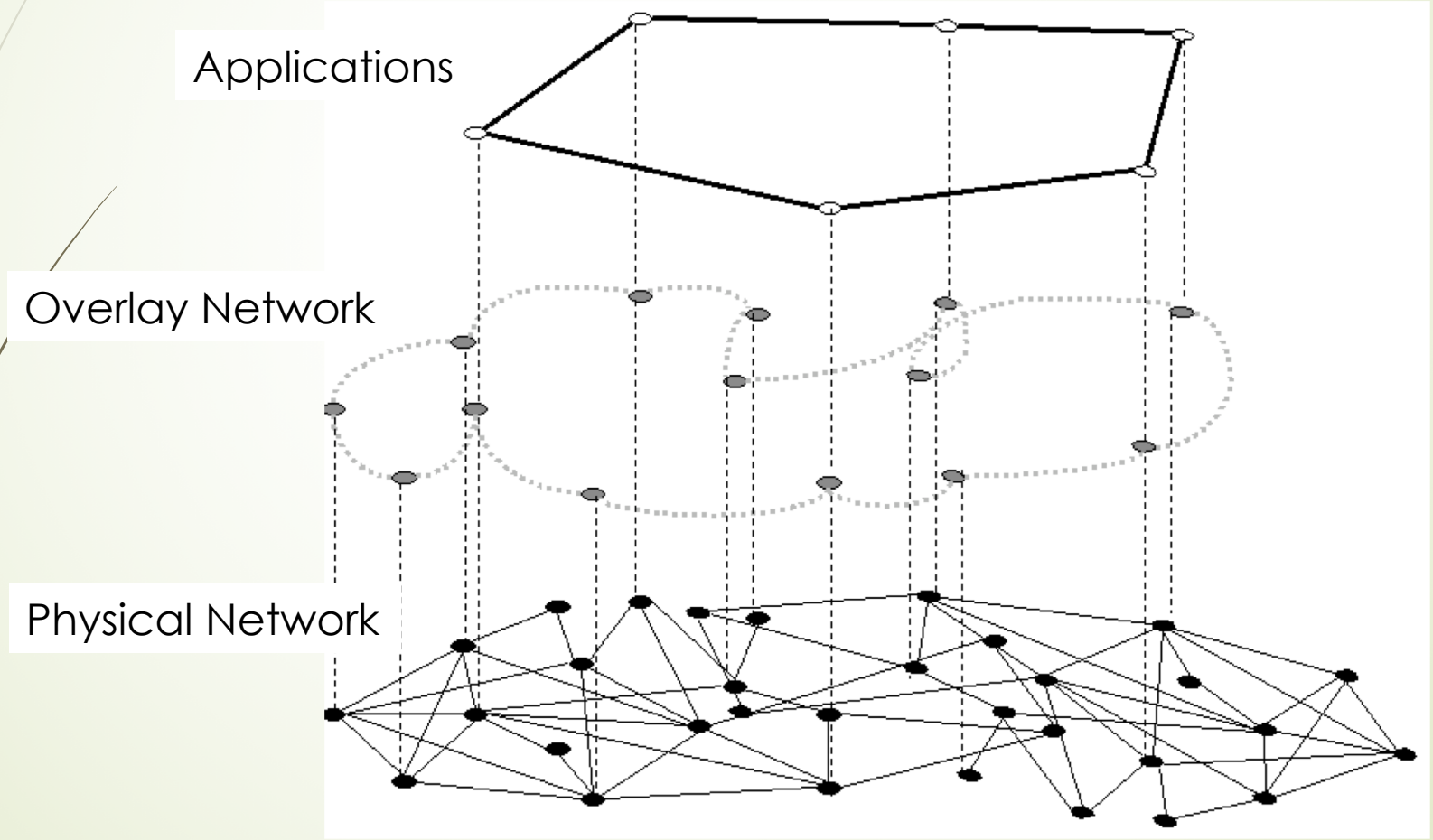
- *no* always-on server
- arbitrary end systems directly communicate
- peers are intermittently connected and change IP addresses

examples:

- file distribution (BitTorrent)
- Streaming (Xunlei Kankan)
- VoIP (Skype)



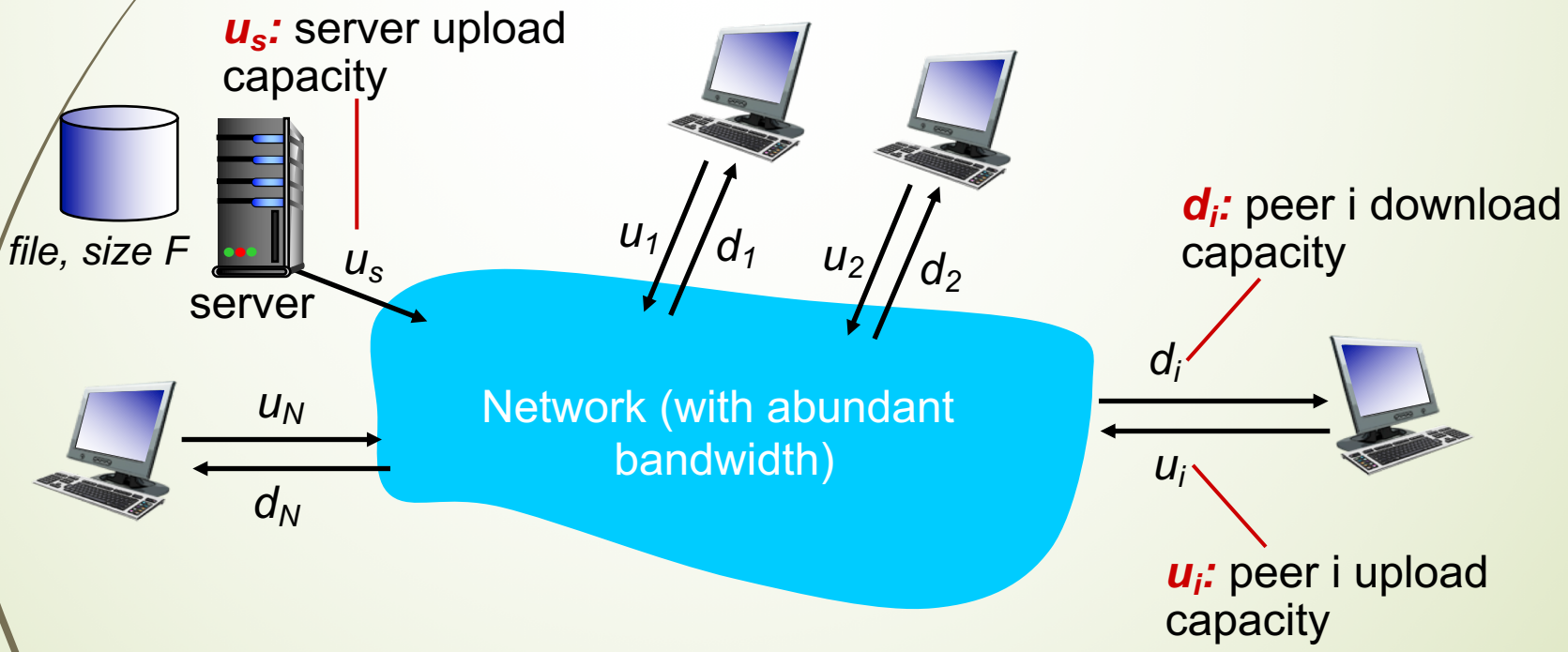
What is the Overlay Network?



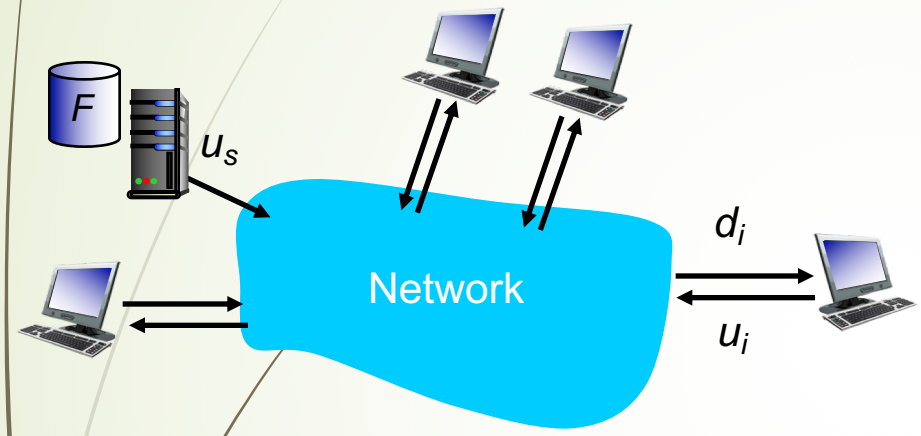
File Dissemination Under Different Architectures

Question: how much time to distribute file (size *F*) from one server to *N* *peers*?

- peer upload/download capacity is limited resource



File Dissemination with Client-Server



- *server transmission:* must sequentially send (upload) *N* file copies:
  - time to send one copy: F/u_s
  - time to send *N* copies: NF/u_s
- *client: each client must download file copy*
  - *dmin = min client download rate*
  - *min client download time: F/dmin*

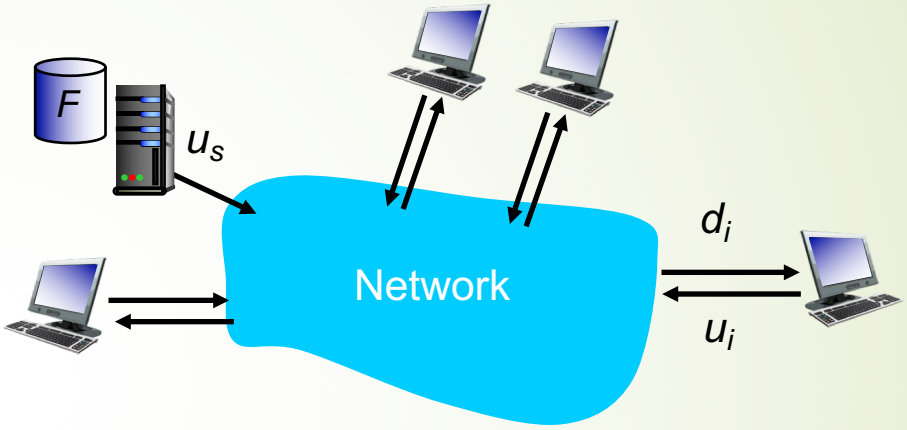
time to distribute F to N clients using client-server approach

$$D_{c\text{-}s} \geq max\{NF/u_s, F/d_{min}\}$$

increases linearly in N

File Dissemination with P2P

- *server transmission:* must upload at least one copy
  - time to send one copy: F/u_s
- *client: each client must download file copy*
  - *min client download time: F/dmin*
- *clients: as aggregate must download NF bits*
  - max upload rate (limiting max download rate) is $u_s + \Sigma u_i$



time to distribute F to N clients using P2P approach

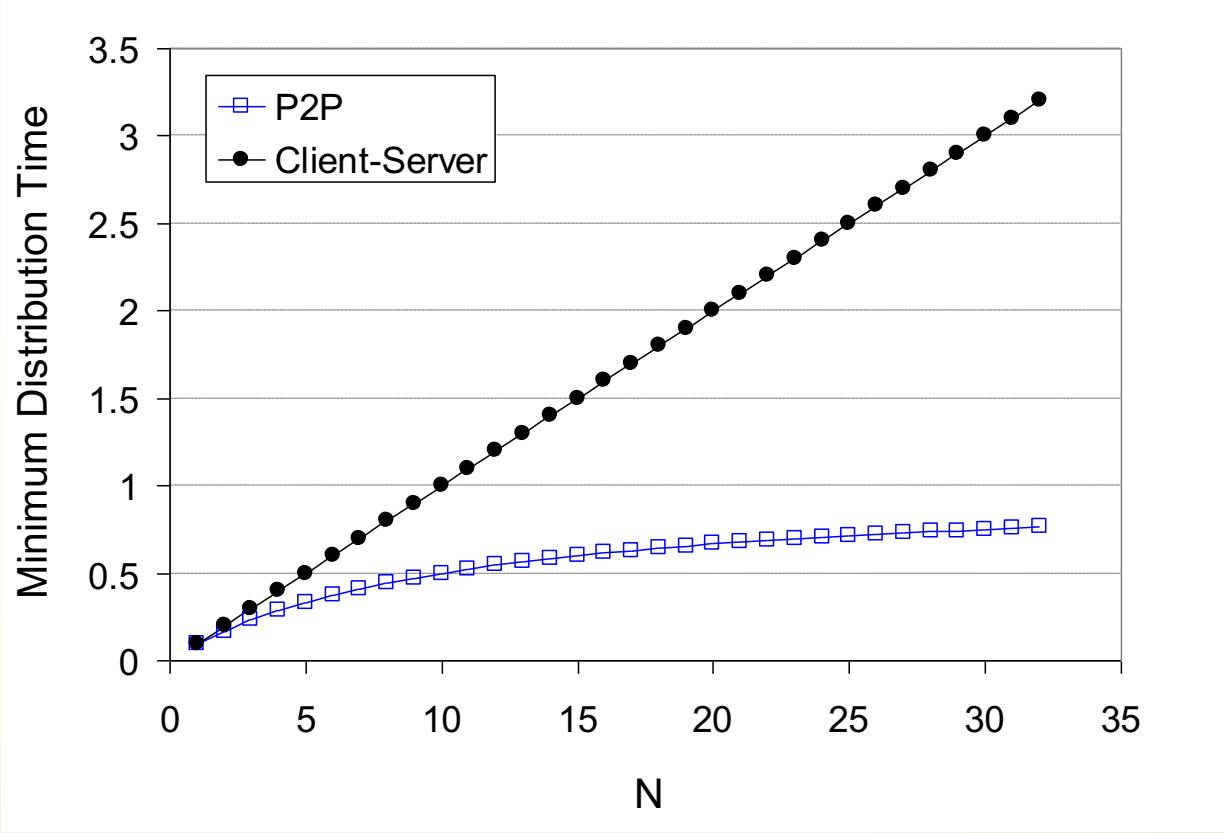
$$D_{P2P} \geq max\{F/u_s, F/d_{min}, NF/(u_s + \Sigma u_i)\}$$

increases linearly in *N* …

… but so does this, as each peer brings service capacity

Sample Comparison Results: Client-Server versus P2P

client upload rate = u, F/u = 1 hour, $u_s = 10u$, $d_{min} \geq u_s$

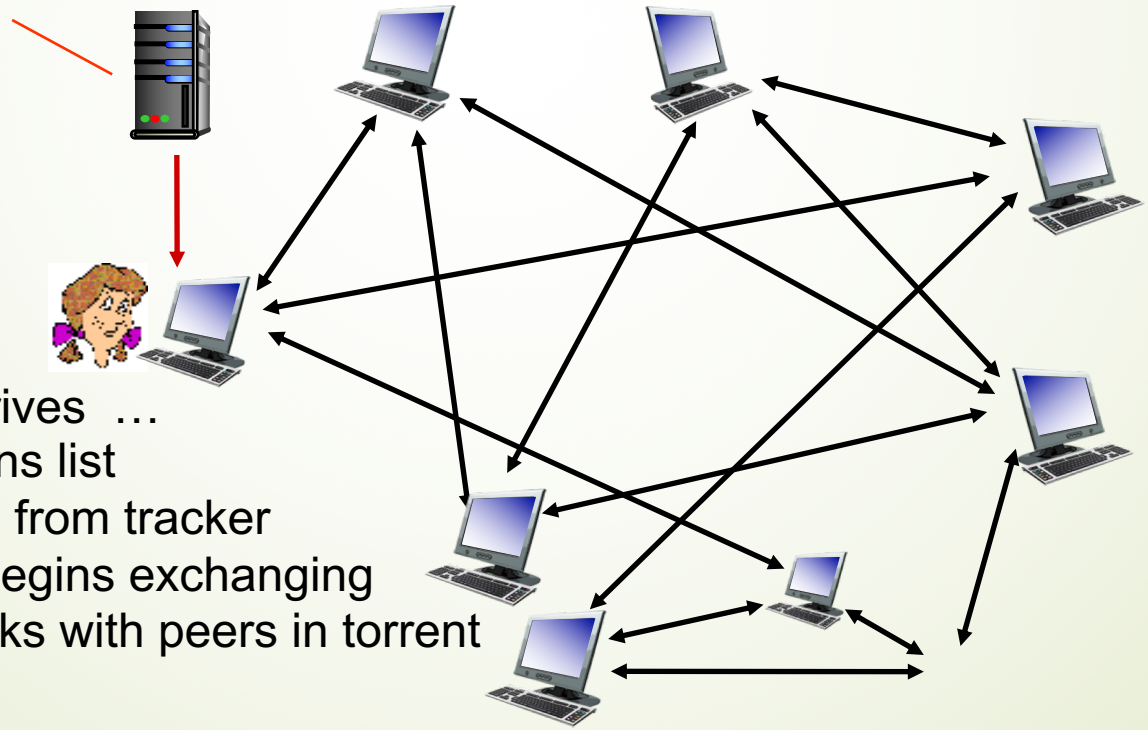


Sample Application: BitTorrent

- file divided into 256Kb chunks
- peers in torrent send/receive file chunks

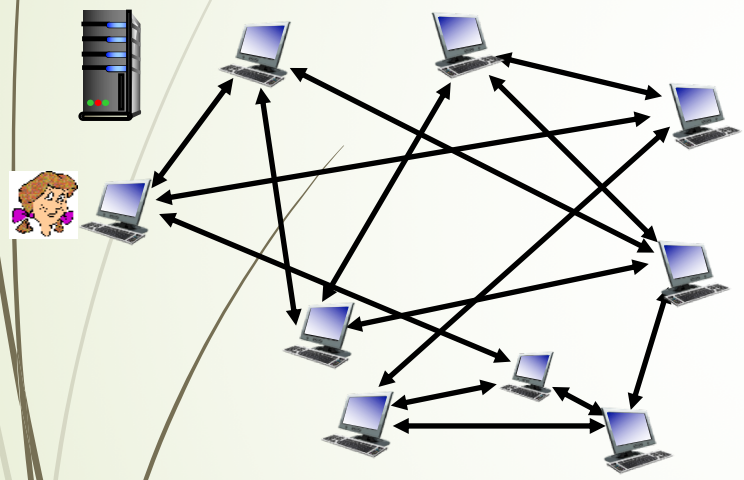
tracker: tracks peers participating in torrent

torrent: group of peers exchanging chunks of a file



Alice arrives …
… obtains list
of peers from tracker
… and begins exchanging
file chunks with peers in torrent

Peers Joining/Leaving BitTorrent



- peer joining torrent:
  - has no chunks, but will accumulate them over time from other peers
  - registers with tracker to get list of peers, connects to subset of peers ("neighbors")
- while downloading, peer uploads chunks to other peers
- peer may change peers with whom it exchanges chunks
- *churn:* peers may come and go
- once peer has entire file, it may (selfishly) leave or (altruistically) remain in torrent

Requesting/Sending Chunks in BitTorrent

requesting chunks:

- at any given time, different peers have different subsets of file chunks
- periodically, Alice asks each peer for list of chunks that they have ← availability map
- Alice requests missing chunks from peers, rarest first

sending chunks: tit-for-tat:

- Alice sends chunks to those four peers currently sending her chunks at highest rate
  - other peers are choked by Alice (do not receive chunks from her)
  - re-evaluate top 4 every 10 secs
- every 30 secs: randomly select another peer, starts sending chunks
  - "optimistically unchoke" this peer
  - newly chosen peer may join top 4

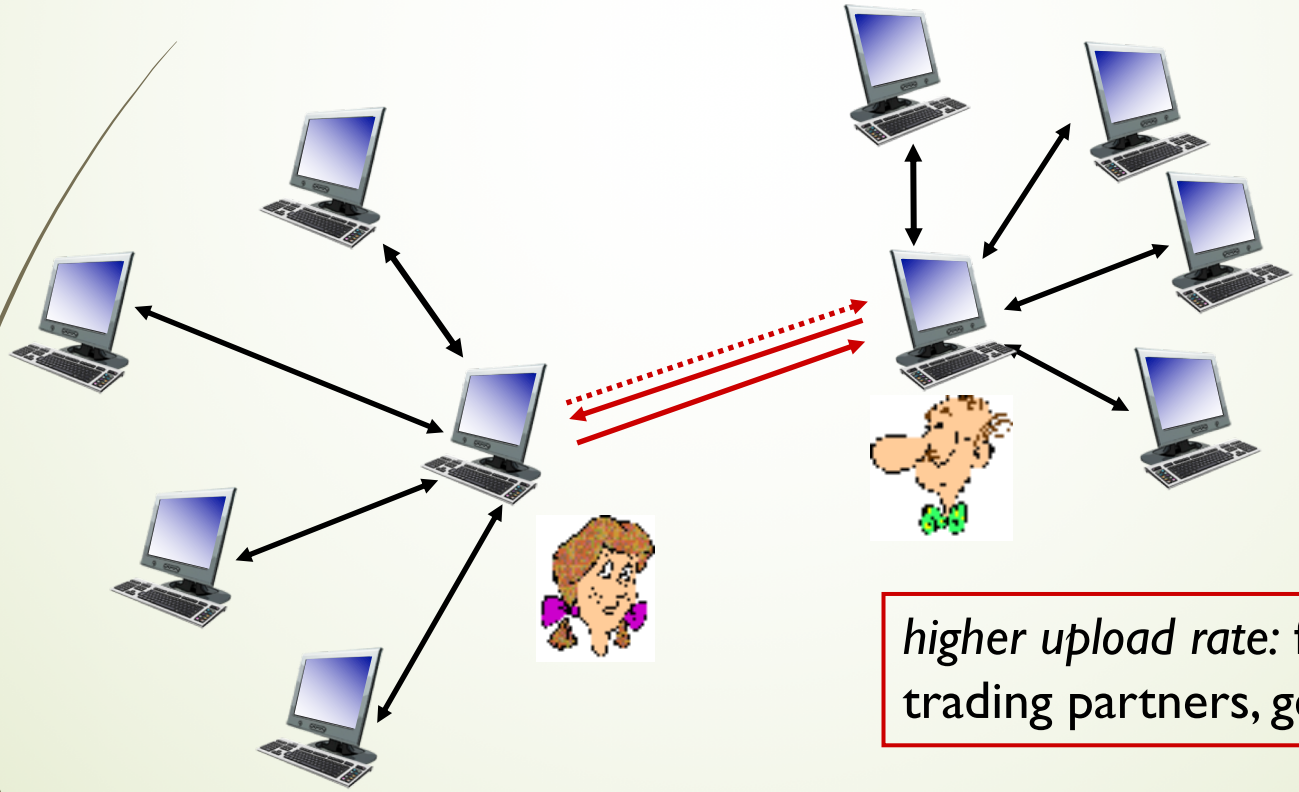
Why do We Need tit-for-tat?

Key of the success of BitTorrent

(1) Alice "optimistically unchokes" Bob

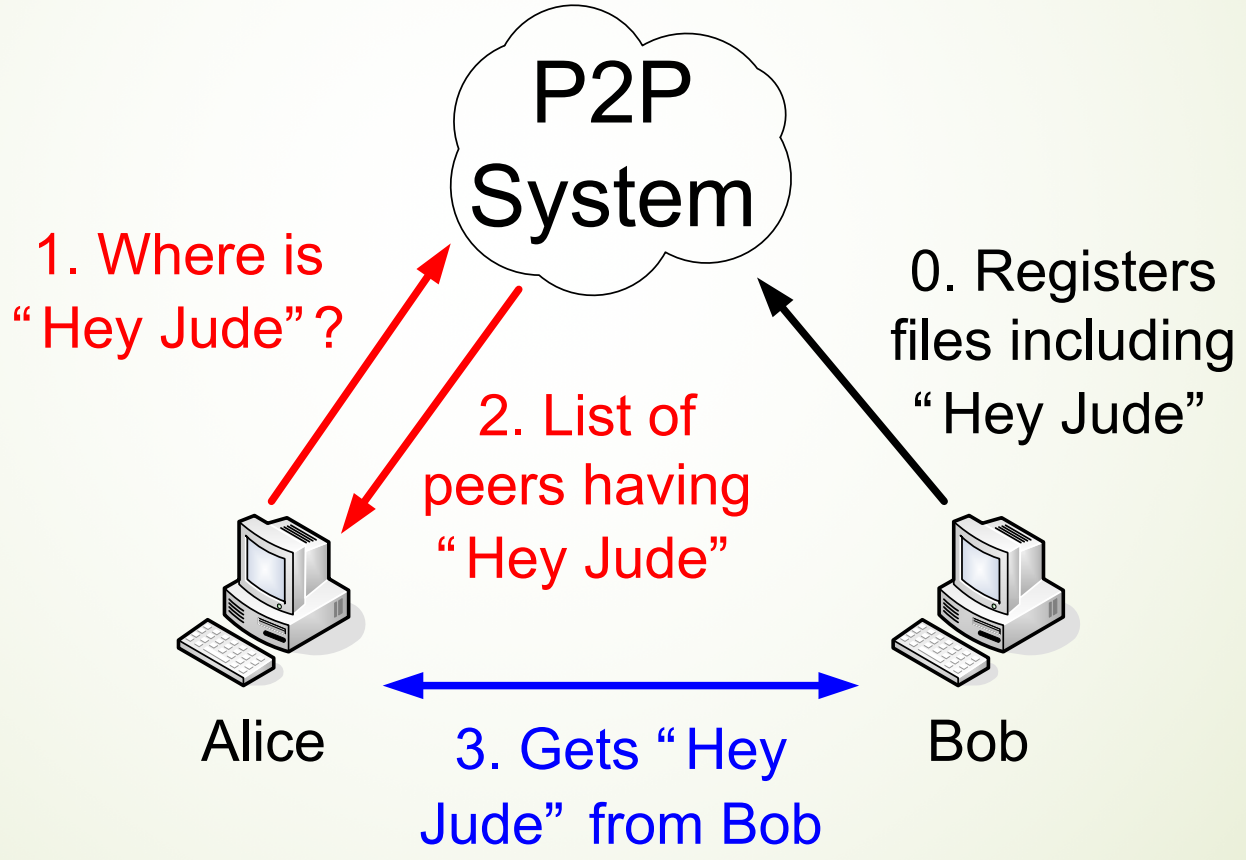
(2) Alice becomes one of Bob's top-four providers; Bob reciprocates

(3) Bob becomes one of Alice's top-four providers



higher upload rate: find better trading partners, get file faster !

Summary: How BitTorrent Work (at Very High Level)



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- Properties of P2P Systems
- Structured and Unstructured P2P
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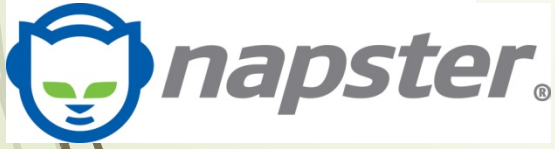
A Partial List of P2P Applications



Classes of P2P Applications

- Search, File Sharing and Content dissemination
  - Napster, Gnutella, Kazaa, eDonkey, BitTorrent
  - Chord, CAN, Pastry/Tapestry, Kademlia,
  - Bullet, SplitStream, CREW, FareCAST
- Communications
  - MSN, Skype, Social Networking Apps
- Storage
  - OceanStore/POND, CFS (Collaborative FileSystems),TotalRecall, FreeNet, Wuala
- Distributed Computing
  - Seti@home
- **Q: Can you think of anything else?**

P2P/Grid Distributed Processing and Crowdsouring

- seti@home
  - Search for Extraterrestrial intelligence
  - Central site collects radio telescope data
  - Data is divided into work chunks of 300 Kbytes
  - User obtains client, which runs in background
  - Peer sets up TCP connection to central computer, downloads chunk
  - Peer does FFT on chunk, uploads results, gets new chunk
- Not P2P communication, but exploit Peer computing power
- Crowdsourcing – Human-oriented P2P

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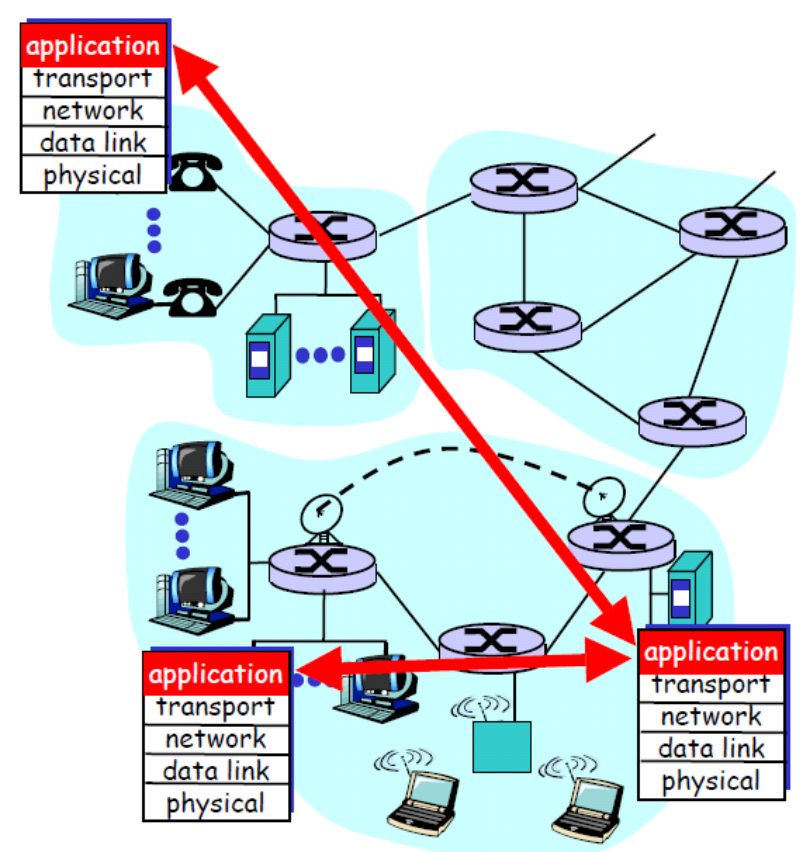
Characteristics of P2P Systems

- Exploit edge resources
  - Storage, content, CPU, Human presence
- Significant autonomy from any centralized authority
  - Each node can act as a Client as well as a Server
- Resources at edge have intermittent connectivity, constantly being added & removed
  - Infrastructure is untrusted and the components are unreliable ← main weakness….

Promising Points of P2P

- Self-organizing
- Massive scalability
- Autonomy: non single point of failure
- Resilience to Denial of Service
- Load distribution
- Resistance to censorship

Revisit Overlay Networks



- **Tremendous design flexibility**
  - Topology, maintenance
  - Message types
  - Protocol
  - Messaging over TCP or UDP
- **Underlying physical network is transparent to developer**
  - But some overlays exploit proximity ← example: reducing RTT among gamers